



ARUNAI ENGINEERING COLLEGE

(An Autonomous Institution)

Velu nagar, Tiruvannamalai-606603

ONV.GL

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

BACHELOR OF ENGINEERING

2024 -2025

FIFTH SEMESTER

**SB8067-Naan Mudhalvan
Laboratory**

(TripAdvisor E-Management)

ARUNAI ENGINEERING COLLEGE

(An Autonomous Institution)

TIRUVANNAMALAI-606 603

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

Certified that this is a bonafide record of work done by

Name

NM ID

University Reg No

Semester

Branch

Year

Staff-in-charge

Head of the Department

Submitted for the

Practical Examination held on

Internal Examiner

External Examiner

TripAdvisor E-Management

INTRODUCTION AND OVERVIEW

The **TripAdvisor E-Management** app, enhanced with Salesforce integration, provides a seamless and efficient platform for travelers to plan, book, and manage every aspect of their journey. This all-in-one travel companion combines the power of Salesforce's CRM tools with TripAdvisor's vast database of reviews, recommendations, and travel services. By leveraging user-generated content and personalized recommendations, it empowers travelers to make more informed decisions and enjoy an optimized travel experience. At the same time, it helps businesses in the travel and hospitality industry better understand customer needs and improve engagement through customized services.

Salesforce is a leading cloud-based Customer Relationship Management (CRM) platform that revolutionizes how businesses interact with their customers. Offering a suite of versatile tools for managing customer relationships, optimizing sales processes, and enhancing customer service, Salesforce has become a cornerstone for businesses across various industries. In particular, the travel industry, which faces constant changes in customer expectations, market trends, and competition, greatly benefits from the advanced capabilities of Salesforce's CRM platform.

OBJECTIVE

Incorporating Salesforce into the operations of a travel business can significantly impact the company's success by achieving specific business goals. These goals are aimed at improving customer satisfaction, operational efficiency, and overall profitability. Below are key business goals that travel businesses aim to achieve by leveraging Salesforce.

Business Goals:

1. Enhance Customer Experience and Loyalty
2. Optimize Sales Processes and Revenue Generation
3. Increase Market Reach and Customer Acquisition
4. Increase Revenue through Upselling and Cross-Selling

Specific Outcomes:

o Streamlined Cross-Department Collaboration o Improved Customer Personalization o Automated Marketing Campaigns with Higher Engagement o 24/7 Self-Service Access for Customers o Enhanced Real-Time Communication with Customers o Faster Issue Resolution and Case Management o Enhanced Compliance and Security

Salesforce Key Features and Concepts Utilized

1. Sales Cloud for lead and opportunity management.
2. Service Cloud for customer service and case management.
3. Commerce Cloud for E-commerce and Booking Management.
4. Community Cloud for Creating Customer and Partner Communities.
5. Salesforce Lightning Experience for Enhanced User Interface and Salesforce Shield for Security and Compliance
6. Marketing Cloud for personalized customer engagement campaigns.
7. Integration with third-party tools using APIs for seamless data synchronization.
8. Automation Tools like Flows, Process Builder, and Workflow Rules.

Explore and In-Depth Overview of implementation

Acceptance Criteria & Solution:

- As the Salesforce User we have to manage the data for the Hotels, Flights, and Food Options for this we have to create some automation for simplification.

- To ensure that when a new Food Option is added or updated, the corresponding Hotel's information is updated accordingly. For example, you might want to maintain a total count of food options for each hotel.
- Also there is automation for the customer benefits if the there buying amount is with respect to some amount then they will get some discounts on their bill
- For the flights there schedule process being involved where the customer who has booked the flight will get the reminder mail alert for knowing proper timing of the flight before 24 hrs it's important to manage the in a good way.
- The system should provide confirmation or notification to the user upon successful sending of the email.

Solution: For the Above requirements of TripAdvisor we have created the solutions by creating the custom objects and Fields the Custom Objects that are created are Hotels, Food Options, Customer & Flights. For the Automation we have used here a flow and triggers and for scheduling the email alerts we have created the Apex Schedulable class so email alerts will be created.

1.Create Object

Hotel Object is created to ensure that when a new Food Option is added or updated with the necessary information

- Enter label : Hotel

- Plural Name : Hotels
- Data Type : (text)
- Field Name : Hotel Name • Click Allow Reports
- Allow Search ? Save

With Above References Create following Object

Food Option ? Data Type ? Auto Number ? Format? FO - {0000}

Flight ? Data Type ? Auto Number ? Format? FL- {0000}

Customer ? Text ? Field Name ? Customer Name

2.Create Fields for Hotel Object

Sr. No.	Field Name	Data Type
1	TotalFoodOptions	Number
2	Date	Date

3.Create Fields For Food Option

Sr. No.	Field Name	Data Type
1	Name	Text

2	Hotel	Hotel(Lookup)
3	Food Amount	Currency

4.Create Fields in the Flight Object

Sr. No.	Field Name	Data Type
1	Name	Date/Time
2	DepartureDateTime	Hotel(Lookup)

5.Create Fields in the Customer Object

Sr. No.	Field Name	Data Type
1	Customer Name	Name
2	Discount Amount	Formula (Currency)
3	Discount Percent	Percentage

Setup > Search Setup

Setup > OBJECT Relationships

Hotel

Fields & Relationships
6 items. Sorted by Field Label

Fields & Relationships

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	lookupUser		
Date	Date_c	Date		
Hotel Name	Name	Text(80)		☒
Last Modified By	LastModifiedById	lookupUser		
Owner	OwnerId	lookupUser (Sharing)		☒
RelationshipOptions	RelationshipOptions_c	Number(18, 0)		

Fields & Relationships

- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Component Layouts
- Field Sets
- Object Cards
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout
- Relationship Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- Validation Rules
- Conditional Field Permissions

Setup > Search Setup

Setup > OBJECT Relationships

Hotel

Fields & Relationships
6 items. Sorted by Field Label

Fields & Relationships

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	lookupUser		
Date	Date_c	Date		
Hotel Name	Name	Text(80)		☒
Last Modified By	LastModifiedById	lookupUser		
Owner	OwnerId	lookupUser (Sharing)		☒
RelationshipOptions	RelationshipOptions_c	Number(18, 0)		

Fields & Relationships

- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Component Layouts
- Field Sets
- Object Cards
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout
- Relationship Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- Validation Rules
- Conditional Field Permissions

Setup Home Object Manager

Search Setup

SETUP - OBJECT MANAGER

Flight

Details

Fields & Relationships

4 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Test History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
ContractEmail	ContractEmail_c	Email		
Created By	CreatedBy	Unrestricted		
DepositedOverTime	DepositedOverTime_c	Date/Time		
Flight Name	Name	Auto Number		✓
Model	Model_c	Lookup (None)		✓
Last Introduced By	LastIntroducedBy	Lookup (User)		
Name	Name_c	Date/Time		
Owner	Owner	Lookup (All Groups)		✓

Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Grouping Rules
Object Access
Triggers
Flow Triggers
Validation Rules

Setup Home Object Manager

Search Setup

SETUP - OBJECT MANAGER

Customer

Details

Fields & Relationships

7 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Test History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Unrestricted		
Customer Name	CustomerName_c	Text (50)		
Customer Name	Name	Text (50)		✓
Discount Amount	DiscountAmount_c	Formula (Currency)		
Discount Percent	DiscountPercent_c	Percent (0-1)		
Last Introduced By	LastIntroducedBy	Lookup (User)		
Owner	Owner	Lookup (All Groups)		✓

Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Grouping Rules
Object Access
Triggers
Flow Triggers
Validation Rules

Create Flow

Create the Flow for the discount for customer when the Amount is greater than 3000 some some Amount of Discounts will be there if the Amount is between 1500 to 3000 so Some Amount of Discount will be there for them

Flow Procedure

Create 3 variable :

Variable > Api name > fold > text > Available for Input

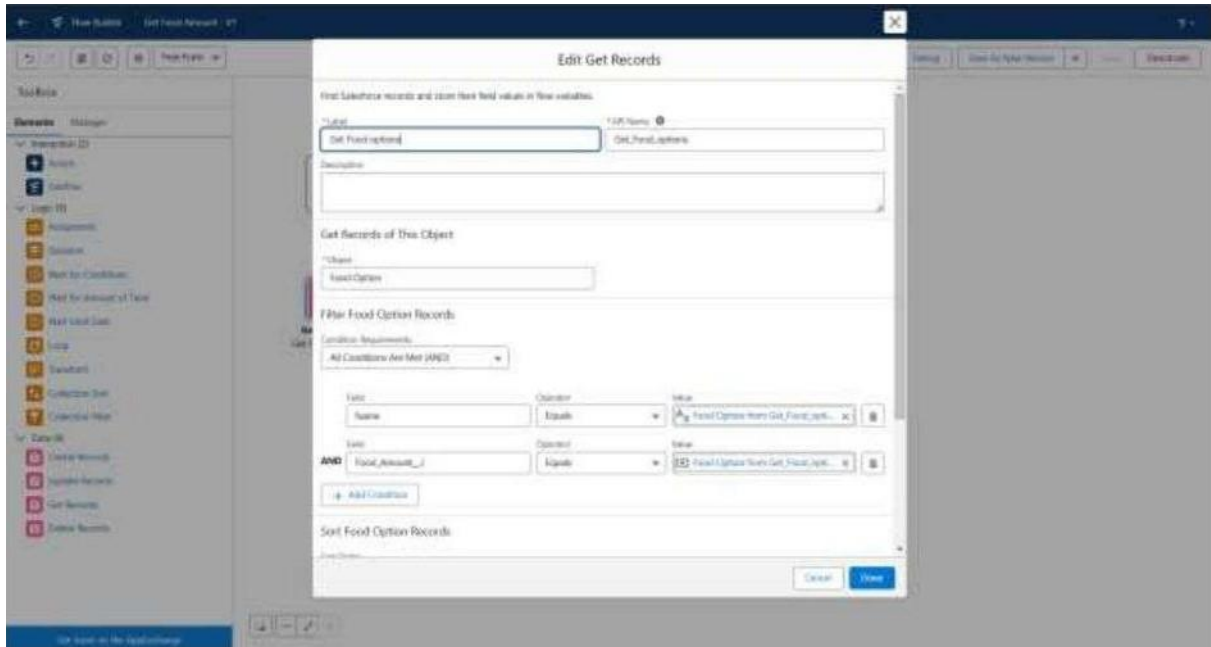
Variable > Api name > csld > text > Available for Input

Variable > Api name > discount > Number



Flow Steps :

1. Get Records
2. Decision Element: Create 2 Outcomes
3. Take the 3 Assignments > Full Discount, Partial Discount & No Discount
4. Update Record Element



Edit Get Records

First Select the records and then filter the values in the variables.

*API Name: *API Name:

Description:

Get Records of This Object

*Object:

Filter Food Option Records

Condition Requirements:

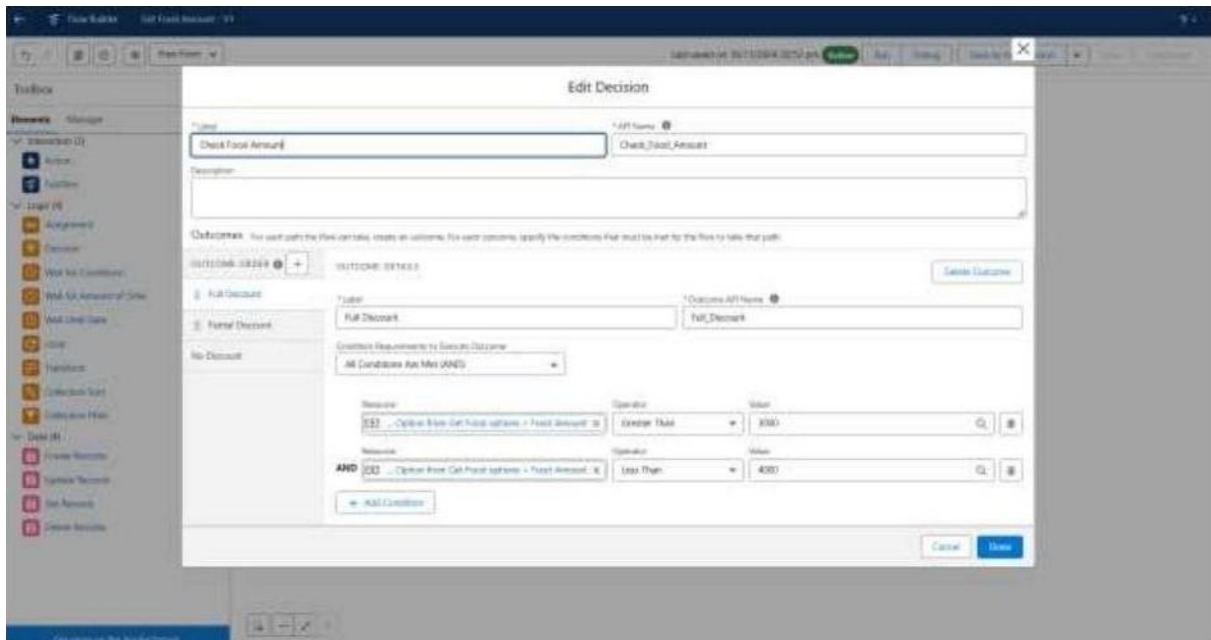
Field	Operator	Value
Food_Amount_1	Equals	Food Option from Get Food Option
Food_Amount_2	Equals	Food Option from Get Food Option

AND

Sort Food Option Records

Sort Order:

Cancel Done



Edit Decision

*API Name: *API Name:

Description:

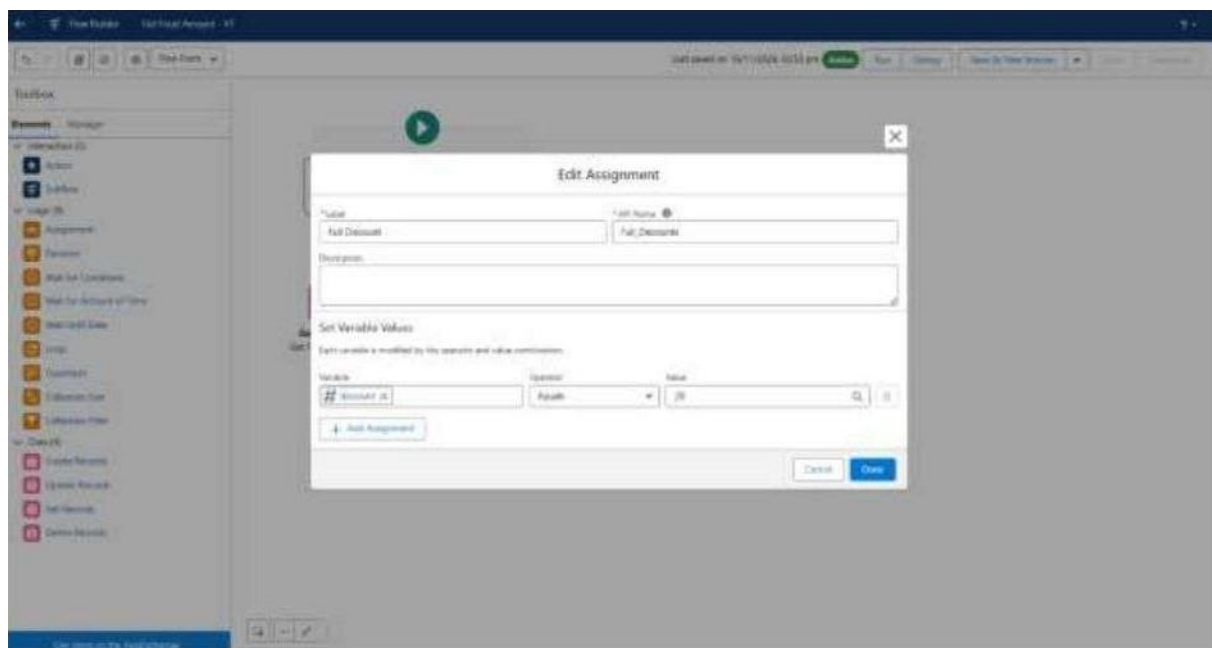
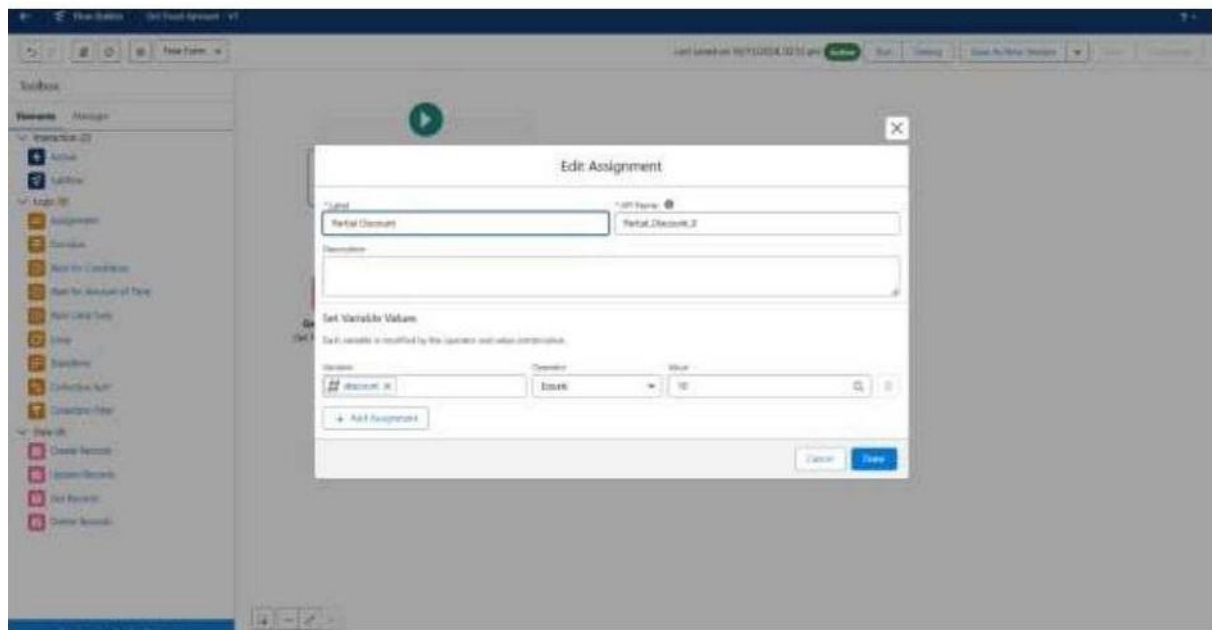
Outcomes: For each path the flow can take, create an outcome. For each outcome, specify the conditions that must be met for the flow to take that path.

OUTCOME DETAILS

Outcome	Condition Requirements to Success Outcome	Value
Full Discount	Food_Amount_1 equals Food Option from Get Food Option	1000
Partial Discount	Food_Amount_1 less than Food Option from Get Food Option	4000

AND

Cancel Done



Edit Assignment

* Label

No Discount

* API Name ⓘ

No_Discount

Description

Set Variable Values

Each variable is modified by the operator and value combination.

Variable

discount ×

Operator

Equals

Value

0



+ Add Assignment

Cancel

Done

Edit Update Records

Update Salesforce records using values from the flow.

* Label

Update_Customer

* API Name ⓘ

Update_Customer

Description

* How to Find Records to Update and Set Their Values

- ☐ Use the IDs and all field values from a record or record collection
- ☒ Specify conditions to identify records, and set fields individually

Update Records of This Object Type

* Object

Customer

Filter Customer Records

Condition Requirements to Update Records

All Conditions Are Met (AND)

Field

Id

Operator

Equals

Value

Id csld ×



+ Add Condition

Set Field Values for the Customer Records

Field

Discount_Percent__c

Value

discount ×



+ Add Field

Cancel

Done

Apex Triggers

Scenario: In the Hotel you have to ensure that when a new Food Option is added or updated, the corresponding Hotel's information is updated accordingly. For example, you might want to maintain a total count of food options for each hotel. To manage the things properly with perspective to the Hotel things should be clearly manageable for making the food options available with respect to hotels

Apex trigger With Handler

Apex Trigger With Handler

```
public class FoodOptionTriggerHandler {  
    // Method to update hotel information based on food options  
    public static void updateHotelInformation(List<Food_Option_c> newFoodOptions,  
List<Food_Option_c> oldFoodOptions, TriggerOperation operation) {  
        Set<Id> hotelIdsToUpdate = new Set<Id>();  
        // Collect unique Hotel Ids affected by food options changes  
        for (Food_Option_c foodOption : newFoodOptions) {  
            hotelIdsToUpdate.add(foodOption.Hotel_c);  
  
            // Update hotel information based on food options  
            List<Hotel_c> hotelsToUpdate = [SELECT Id, Name, TotalFoodOptions c FROM Hotel  
WHERE Id IN :hotelIdsToUpdate]; for  
            (Hotel_c hotel : hotelsToUpdate) {  
  
                // Recalculate total food options count  
                Integer totalFoodOptions = [SELECT COUNT() FROM Food_Option_c WHERE Hotel  
:hotel.Id];          hotel.TotalFoodOptions_c          =  
                totalFoodOptions;  
  
                // Update hotels with new total food options count update  
                hotelsToUpdate;
```

```

1 public class FoodOptionTriggerHandler {
2
3     // Method to update hotel information based on food options
4
5     public static void updateHotelInformation(List<Food_Option__c> newFoodOptions, List<Food_Option__c> oldFoodOptions, TriggerOperation operation) {
6
7         Set<Id> hotelIdsToUpdate = new Set<Id>();
8
9
10        // Collect unique Hotel Ids affected by food options changes
11
12        for (Food_Option__c foodOption : newFoodOptions) {
13
14            hotelIdsToUpdate.add(foodOption.Hotel__c);
15
16        }
17
18        // Update hotel information based on food options
19
20        List<Hotel__c> hotelsToUpdate = [SELECT Id, Name, TotalFoodOptions__c FROM Hotel__c WHERE Id IN :hotelIdsToUpdate];
21
22
23    }
24 }
  
```

Trigger

```

trigger FoodOptionTrigger on Food_Option__c (after insert, after update, after ) {
    if(trigger.isInsert && trigger.isAfter){
        FoodOptionTriggerHandler.updateHotelInformation(trigger.new);
    }
}
  
```

Apex Schedule

Create the Reminder mail for the customer who has booked the flight according to that booking set the Apex schedule so mail will be sent prior to 24hrs.

Note: Please create the required field for Scheduled Apex Code

Apex Schedule Class Solution

```

public class FlightReminderScheduledJob implements Schedulable {
  }
  
```

```
public void execute(SchedulableContext sc) {  
    sendFlightReminders();
```

```
private void sendFlightReminders() {
```

```
// Query for flights departing within the next 24 hours
```

```
    List<Flight_c> upcomingFlights = [SELECT Id, Name, DepartureDateTime c FROM Flight  
                                     WHERE DepartureDateTime c :DateTime.now()  
                                     AND DepartureDateTime_c :DateTime.now().addDays(1)]; for  
(Flight c flight : upcomingFlights) {
```

```
    // Customize the logic to send reminder emails
```

```
    // For this example, we'll print a log message; replace this with your email sending logic.
```

```
    System.debug('Sending reminder email for Flight ' + flight.Name + ' to ' +  
flight.ContactEmail_c);
```

```
// Example: Send email using Messaging.SingleEmailMessage
```

```
    Messaging.SingleEmailMessage email = new Messaging.SingleEmailMessage();  
    email.setToAddresses(new List<String>{ flight.ContactEmail_c });  
    email.setSubject('Flight Reminder: ' + flight.Name); email.setPlainTextBody('This is a  
reminder for your upcoming flight ' + flight.Name +  
    ' departing on ' + flight.DepartureDateTime_c);  
    Messaging.sendEmail(new List<Messaging.SingleEmailMessage>{ email });
```



```

1 * public class FlightReminderScheduledJob implements Schedulable {
2
3 *   public void execute(SchedulableContext sc) {
4       sendFlightReminders();
5   }
6
7 *   private void sendFlightReminders() {
8       // Query for flights departing within the next 24 hours
9 *       List<Flight__c> upcomingFlights = [SELECT Id, Name, DepartureDateTime__c, ContactEmail__c
10                                           FROM Flight__c
11                                           WHERE DepartureDateTime__c >= :Date.today()
12                                           AND DepartureDateTime__c <= :Date.today().addDays(1)];
13
14
15 *       for (Flight__c flight : upcomingFlights) {
16           // Customize the logic to send reminder emails
17           // For this example, we'll print a log message, replace this with your email sending logic.
18           System.debug('Sending reminder email for Flight ' + flight.Name + ' to ' + flight.ContactEmail__c);
19
20           // Example: Send email using Messaging.SingleEmailMessage
21           Messaging.SingleEmailMessage email = new Messaging.SingleEmailMessage();
22           email.setToAddresses(new List<String>{ flight.ContactEmail__c });
23           email.setSubject('Flight Reminder: ' + flight.Name);
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100

```

The FlightReminderScheduledJob class implements the Schedulable interface, and the execute method is where you put the logic to send reminder emails.

The sendFlightReminders method queries for flights departing within the next 24 hours. You can customize the query based on your specific requirements.

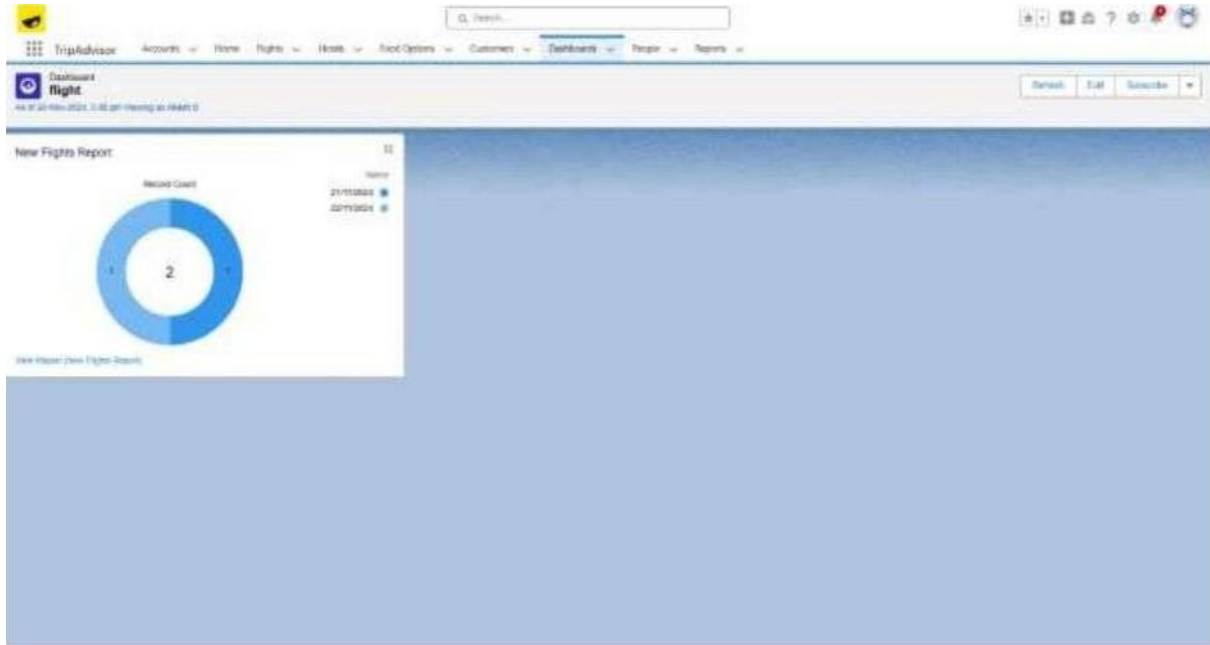
Create the Apex code in an anonymous Window to execute the Apex Code

// Schedule the job to run every day at a specific time (e.g., 6 AM)

String cronExp = '0 06 * * ?';

System.schedule('FlightReminderJob', cronExp, new FlightReminderScheduledJob());

PROJECT OUTPUT:



The screenshot shows the TripAdvisor Flights page. The top navigation bar includes links for Accounts, Home, Flights, Hotels, Food & Drink, Customers, Dashboards, People, and Reports. The main content area displays a 'Recently Viewed' section with a table of flights. The table has columns for 'Flight Name', 'From', 'To', and 'Price'. The table contains two rows of data. The first row shows a flight from 'P1' to 'P2' with a price of 'USD 100.00'. The second row shows a flight from 'P1' to 'P2' with a price of 'USD 100.00'. The table is labeled 'Recently Viewed' and 'Flights'.

	Flight Name	From	To	Price
1	P1 - P2	P1	P2	USD 100.00
2	P1 - P2	P1	P2	USD 100.00

TipAdvisor Accounts Home Rights Work Recent Updates Customers Dashboards People Reports

Search

Recently Viewed

2 items - Updated a minute ago

	Host Name
1	Demo Host
2	Demo Host

TipAdvisor Accounts Home Rights Work Recent Updates Customers Dashboards People Reports

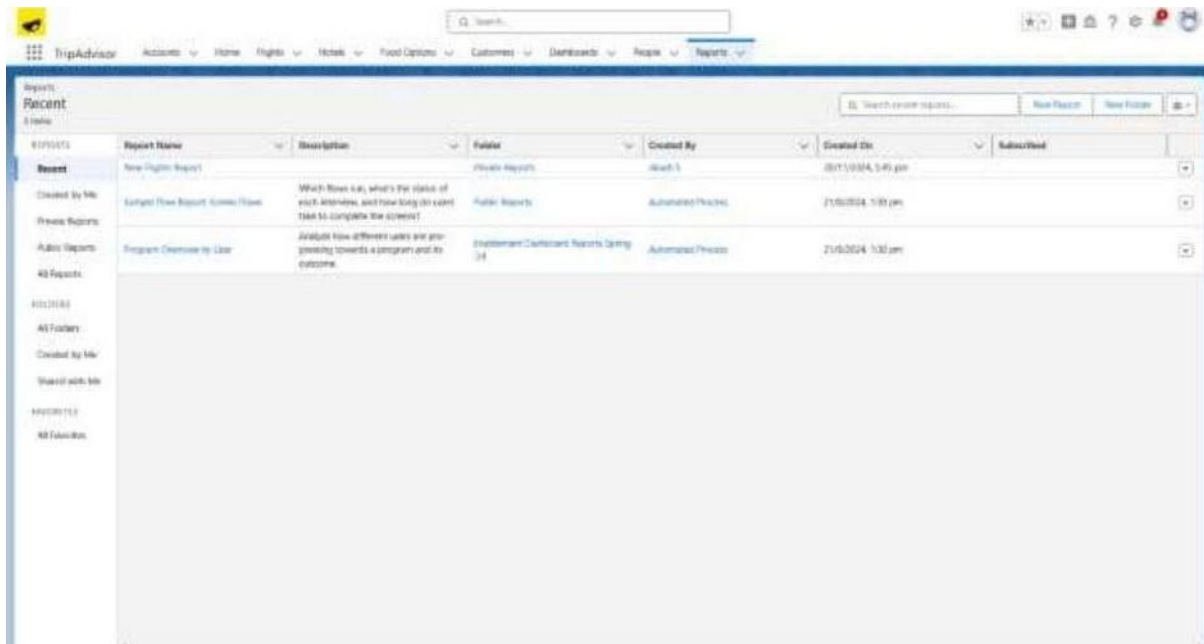
Search

Customers

Recently Viewed

2 items - Updated a few seconds ago

	Customer Name
1	Hardware
2	Network



Testing and Quality Assurance

- **Functional Testing:** Ensure that all features, tools, and functions implemented within Salesforce (e.g., Sales Cloud, Service Cloud, Marketing Cloud) are working as expected.
- **Data Integrity Testing:** Verify the accuracy and consistency of the data across all integrated platforms.
- **Security Testing:** Ensure that all customer and business data is secured and compliant with regulations (e.g., GDPR).
- **Usability Testing:** Ensure that the user interface (UI) is intuitive and meets the needs of both customers and internal teams.
- **Performance Testing:** Ensure that the platform can handle a high volume of users, transactions, and data without performance degradation.

Critical Use Cases and Solutions Delivered by Salesforce in the Implementation

1. Streamlining Customer Booking and Reservation Process

Challenge: Manual and fragmented booking processes leading to delays and errors.

Solution: Salesforce's Commerce Cloud and Sales Cloud integrate booking systems, seamless online reservations, real-time availability checks, and smooth payment processing.

2. Improving Customer Service and Case Management

Challenge: Slow and inefficient handling of customer service inquiries, resulting in lower satisfaction.

Solution: Salesforce Service Cloud automates case management workflows, provides a 360-degree customer view, and integrates chatbots to resolve issues faster, improving response time and service quality.

3. Personalized Marketing and Customer Engagement

Challenge: Generic marketing campaigns not tailored to individual preferences, leading to lower engagement.

Solution: Marketing Cloud uses customer data and behavior analytics to create personalized campaigns, send tailored offers, and optimize engagement through email, social, and web.

4. Optimizing Sales and Lead Management

Challenge: Disorganized lead management and difficulty tracking sales opportunities across teams.

Solution: Sales Cloud enables efficient lead tracking, pipeline management, and collaboration tools to help sales teams prioritize opportunities and close deals faster.

5. Real-Time Analytics for Business Insights

Challenge: Lack of actionable insights, making it difficult to adjust business strategies in real-time.

Solution: Einstein Analytics delivers AI-powered insights, creating detailed reports and dashboards that provide key performance metrics, predictive trends, and realtime decision-making support.

6. Automating Customer Feedback and Review Management

Challenge: Manual collection and management of customer reviews, leading to inefficiency.

Solution: Salesforce integrates review collection tools and automates follow-ups with customers, leveraging Marketing Cloud and Service Cloud to streamline feedback loops and improve customer experience.

7. Streamlining Partner and Supplier Communication

Challenge: Difficulties in managing communication with external travel partners, leading to delays and mismanagement of data.

Solution: Experience Cloud enables seamless communication between the business and its partners through secure, branded portals where partners can access real-time data, updates, and documents.

8. Integrating External Travel Tools and Systems

Challenge: Inconsistent data and communication with third-party systems (e.g., hotel booking, flight booking).

Solution: Salesforce's API integration capabilities ensure seamless data exchange and synchronization with external travel platforms, improving operational efficiency and reducing manual entry errors.

9. Improving Operational Efficiency through Workflow Automation

Challenge: Time-consuming manual tasks leading to inefficiencies in operations.

Solution: Automation tools such as Process Builder, Flows, and Workflow Rules automate routine tasks like booking confirmations, invoice generation, and follow-up emails, reducing manual effort and errors.

10. Scalability and Flexibility to Handle Growing Customer Data

Challenge: The need to scale operations as the customer base grows, without compromising on performance or data management.

Solution: Salesforce's cloud architecture allows for scalable data storage, handling increasing customer interactions and maintaining high performance even as the business grows.

Conclusion:

The TripAdvisor E-Management Project showcases how Salesforce can address key challenges in the travel industry. By utilizing Salesforce's tools, the project has created an efficient, customer-focused platform. It streamlines operations and automates processes, improving response times and customer service. Real-time data integration enhances decision-making and business insights. The platform provides personalized experiences for customers, boosting engagement. With robust security measures, the system ensures data protection and compliance. The project empowers TripAdvisor to scale and grow in a competitive market. Ultimately, Salesforce helps drive business success through enhanced operational efficiency and customer satisfaction.